© 2025 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works. Accepted manuscript of: T. Yan, J. Chen, S. Yao, X. Duan, Y. Wang, H. Wang, and G. Chen, "Design of a Novel Force-Controlled End-Effector With Passive Structural Compliance and Intrinsic Contact Sensing," IEEE Transactions on Automation Science and Engineering, vol. 22, pp. 23202–23213, 2025. DOI: 10.1109/TASE.2025.3627234

Design of a Novel Force-Controlled End-effector With Passive Structural Compliance and Intrinsic Contact Sensing

Tianyi Yan, Jianhuan Chen, Siyue Yao, Xuyang Duan, Yanjun Wang, Hao Wang, Genliang Chen

***Abstract*—This paper presents a novel active force-controlled end-effector with passive structural compliance and intrinsic contact sensing capabilities. Slender elastic beams are introduced to provide the end-effector with structural compliance to accommodate inevitable fluctuations through deformation during operation. Meanwhile, flexible strain gauges are embedded into the slender elastic beams to endow the end-effector with intrinsic sensing capability. A kinetostatic model is established in closed-form to map the local deflections of strain gauges to the contact force with the environment. On this basis, model-based feedback control can be readily implemented to actively adjust the contact force between the end-effector and the workpiece. A prototype is developed, on which a variety of experiments are conducted to validate the effectiveness of the proposed end-effector. The results demonstrate that the end-effector can control the contact force accurately, with a maximum error of around 0.65*N*. Furthermore, a demonstration of robotic grinding using the developed prototype showcases its potential for practical applications.**

***Note to Practitioners*—This paper is motivated by the demand for intelligent force-controlled end-effectors in robotic manufacturing and interactive scenarios. In this work, a novel force-controlled end-effector for robotic grinding is presented. Slender elastic beams are utilized to output force and enable the end-effector with structural compliance. By integrating flexible strain gauges into the slender elastic beams, the end-effector achieves intrinsic contact sensing with the environment. Its structural compliance allows the end-effector to adapt to fluctuations in robot motion during the manufacturing process. The intrinsic force-sensing capabilities enable precise feedback control, critical for maintaining high manufacturing quality. The compact design of the end-effector improves its adaptability and facilitates easy installation in a variety of settings. Future research will focus on achieving more degrees of freedom in orientation/force hybrid control. Therefore, the presented end-effector has the potential for applications in robotic manufacturing, including tasks requiring compliant manipulation and precise force control, as well as human-robot collaboration environments.**

***Index Terms*—Force-controlled end-effector, structural compliance, contact sensing, robotic grinding.**

This work was supported in part by major projects of the Ministry of Industry and Information Technology of China under Grants CBG01N23-04-03, and by the Natural Science Foundation of China (NFSC) under the Grants 52022056. (Corrsponding author: Genliang Chen.)

Tianyi Yan, Jianhuan Chen and Siyue Yao are with the Shanghai Key Laboratory of Intelligent Robotics, Meta Robotics Institute, Shanghai Jiao Tong University, 200240, Shanghai, China (e-mail: yantianyi@sjtu.edu.cn; triumph_cjh@sjtu.edu.cn; sy.yao@sjtu.edu.cn).

Xuyang Duan and Genliang Chen are with the Shanghai Key Laboratory of Intelligent Robotics, Meta Robotics Institute and the State Key Laboratory of Mechanical System and Vibration, Shanghai Jiao Tong University, 200240, Shanghai, China (e-mail: clf@sjtu.edu.cn; leungchan@sjtu.edu.cn).

Hao Wang is with the State Key Laboratory of Mechanical System and Vibration and the Shanghai Key Laboratory of Digital Manufacturing for Thin-Walled Structures, Shanghai Jiao Tong University, 200240, Shanghai, China (e-mail: wanghao@sjtu.edu.cn).

Yanjun Wang is with the institute of Marine Equipment, Shanghai Jiao Tong University, Shanghai, 200240, P.R. China (e-mail: yanjunwang@sjtu.edu.cn).

I. Introduction

In the field of manufacturing, particularly within the domain of surface finishing, the capacity to exert precise control over contact forces is pivotal for assuring high surface quality. Traditional manual operations provide experience-based force control but are hindered by work stability issues and elevated labor costs [1] [2]. In contrast, employing computer numerically controlled (CNC) machine tools or robotic arms enhances operational flexibility and positional accuracy, yet these technologies often fall short in terms of precise force control [3] [4] [5]. Therefore, integrating end-effectors at the terminal end of robotic arms emerges as a novel solution that balances flexible operation with precise force control [6].

Current developments in force-controlled end-effectors are classifies into three primary types. The first type utilizes constant force mechanisms (CFMs) to maintain a steady force output, thus eliminating the need for active control. Chen and Lan [7] introduce an adjustable constant-force mechanism (ACFM) that combines the negative stiffness of a bistable mechanism with the positive stiffness of a linear spring, enabling the generation of a constant force output. Zhang et al. [8] explore the second bending mode of a fixed-guided compliant beam to develop a compliant CFM, validated on large-aperture optical reflective mirrors with freeform surfaces. Building upon the integration of negative and positive stiffness mechanisms, Xu and Wei [9] conduct extensive studies to implement CFMs in robotic end-effectors for polishing experiments, which reduce force overshoot and enhance force accuracy. Further advancing this technology, Wu et al. [10] refine the constant force's magnitude by adjusting the preloading displacement of the positive-stiffness mechanism and introduce counterweights to mitigate the effects of gravity, ensuring consistent force application even at wide tilt angles of the end-effector.

The second type involves active end-effectors that incorporate sensors and control algorithms, enabling real-time, dynamic adjustments to force outputs. Bone and Elbestawi [11] pioneer this by deriving design objectives from a dynamic analysis of the combined system of a robot arm and the deburring process dynamics, resulting in a precision-enhancing end-effector for robotic deburring. Similarly, Ma et al. [12] [13] develop a three-DOF end-effector equipped with position-based force/impedance control algorithms that ensure rapid response and high resolution in both position and force control. Mohammad et al. [14] utilize feedback from force sensors to regulate a linear hollow voice coil actuator, effectively reducing vibrations and achieving force tracking. Fan et al. [15] introduce a smart end-effector designed for the specific challenges of polishing thin-walled blisks. It features active

contact force force control and vibration suppression using a gravity-compensated force controller and an eddy current damper.

The third type is a hybrid solution that combines the compliance components with the active control systems. Wang et al. [16] [17] develop an end-effector with a parallel configuration, using four-bar linkages and pneumatic artificial muscles for antagonistic force control. Bao et al. [18] create a constant-force end-effector based on a parallel, motor-spring architecture, incorporating a displacement measurement unit for real-time force adjustments during abdominal ultrasonography. Qin et al. [19] introduce a compliant end-effector for robotic polishing, equipped with flexible struts and a capacitive position sensor to measure displacements and estimate contact forces. Additionally, Chen et al. [20] develop an active compliant force force-controlled end-effector for grinding curved parts, analyzing the impact of stiffness in elastic components and grinding tools on force control.

While active end-effectors are capable of achieving real-time control of contact forces, their lack of compliance makes it difficult to avoid large force overshoots and excessive contact with the workpiece. Although the use of constant force mechanism endows the end-effector with compliance, they do not allow for real-time adjustments to contact force, only permitting the output of a preset specified force. Regarding the hybrid solution that combines compliance components with an active control system, it still relies on sensors external to the end-effector to control the contact force, lacking intrinsic sensing capabilities for the contact force.

To overcome existing deficiencies, a newly designed force-controlled end-effector is proposed in this paper by integrating passive structural compliance and intrinsic contact sensing, as depicted in Fig. 1. Slender elastic beams are employed to impart compliance through structural deformation while simultaneously generating force. Additionally, flexible strain gauges embedded into these beams measure local deformations, equipping the end-effector with capabilities for contact force sensing. A discretization-based approach for tackling large deflection problems of elastic beams is utilized to derive the kinetostatic and sensing model. Consequently, the end-effector proposed in this paper boasts a compact and simple structure that not only provides compliance but also supports real-time intrinsic sensing and control of contact forces. A prototype is developed and tested, demonstrating its capability to control and sense contact forces and accomplish the grinding tasks.

The rest of this paper is organized as follows: Section II exhibits the design and development of the proposed end-effector. Then in Section III, the kinetostatic model of the end-effector is derived in closed-form using a discretization-based approach. Section IV introduces a deflection sensing methodology through strain gauges and establish the sensing model. To validate the performance of the proposed design, Section V presents the prototyping of the end-effector, and various experiments are conducted to verify the performance and demonstrate its application in robotic grinding. Finally, conclusions are drawn in Section VI.

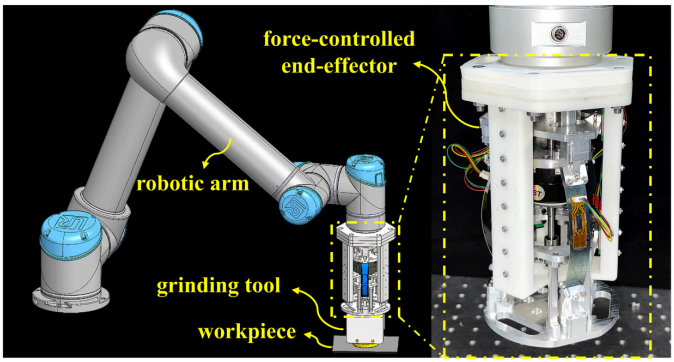


Fig. 1. The developed force-controlled end-effector

II. Mechanism Design and Prototype Fabrication

A. Mechanism design

The detailed architecture of the developed end-effector is illustrated in Fig .2. Central to the design is a motor equipped with a leadscrew, located in the middle part of the end-effector and connected to a moving platform. This platform glides along three internally spaced guide rails at 120-degree intervals using sliders. The end platform is connected to the middle platform, which also slides along these guide rails. This configuration is specifically designed to minimize the total length of the end-effector, enhancing its compactness.

The force-generating unit of the end-effector consists of slender elastic beams, arranged at 120-degree intervals. Notably, these beams are pre-deformed form their original straight configuration and are connected to the platforms via connectors angled at 30 degrees. Flexible strain gauges, integrated into flexible printed circuit board (FPCB), are embedded within the elastic beams to measure local deflections. This setup enables the end-effector to sense and control contact forces based on sensory feedback, providing both passive structural compliance and force sensing capabilities.

Additionally, the introduction of slender elastic beams reduces the stiffness in the horizontal direction, rendering it more vulnerable to disturbances during vertical movements. To mitigate this, a linear bearing is positioned at the bottom of the frame. The end of the leadscrew passes through this bearing, functioning as a constraint. This setup ensures that the moving platform, propelled by the leadscrew, moves precisely in the vertical direction, thereby maintaining a stable output force in this orientation.

B. Prototype fabrication

The prototype of the proposed end-effector utilizes easy-to-access materials and simple fabrication techniques. The frame are made of 3*D*-printed plastic to minimize weight. The moving platform and connectors of slender elastic beams are fabricated from aluminum alloy to prevent deformation under load while satisfying lightweight requirements. The elastic beams are made of spring steel characterized by an elastic modulus of 196 *GPa* and a shear modulus of 79.5 *GPa*. These beams are treated as homogeneous materials with a length of 80 mm and a cross-sectional area of 14mm by 0.25 mm. Each

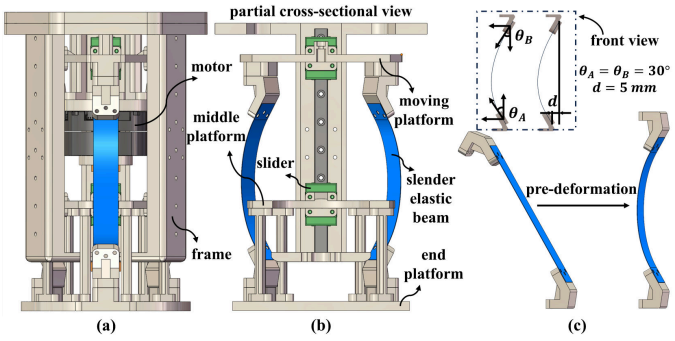


Fig. 2. Architecture of the proposed end-effector. (a) Overall structure, (b) Partial cross-sectional view, (c) Parameters of slender elastic beam

beam is equipped with a flexible strain gauge integrated with FPCB as the deflection sensor, affixed to the beam with the adhesive. The size and weight parameters of the end-effector are detailed in Table I, demonstrating a compact size and light weight.

TABLE I
Size and weight parameters of the proposed end-effector

Length (mm)	Width (mm)	Height (mm)	Weight (g)
100	100	154	697.2

III. Kinetostatics Modeling and Analysis

This section presents the kinetostatic modeling and analysis of the proposed end-effector using a discretization method for large deformation problems of elastic beams. Then, the output force of the developed end-effector can be determined efficiently in terms of the given inputs.

A. Kinetostatics modeling

As illustrated in Fig. 2, the initial state of the elastic beam is obtained by pre-deforming a straight beam. Consequently, this initial state presents a bending configuration characterized by a specific angle, which is depicted in the front view. A discretization-based method, previously developed for addressing large deflection issues in flexible links [21], is applied. By employing the principle axes decomposition [22], the elastic beam is segmented into numerous small segments. Each segment incorporates a 6-DOF linkage, consisting of rigid bodies connected by elastic joints. This modeling approach, which transforms continuum structural mechanics into a rigid, hyper-redundant multibody system, enables the application of established robotics kinematics and statics principles to analyze complex large deformation problems in continuum structures [23]. Given the minimal thickness relative to the width and length of the slender elastic beams employed in our study, the analysis will focus primarily on the bending and torsional terms, disregarding internal shear and compression forces [24][25]. Considering the fixation method and layout of the elastic beam, torsion is negligible during the operation of the end-effector. Hence, only the bending terms need to be considered in this study, which means that the deformation process of a single beam can be regarded as a planar problem, as shown in Fig. 3(a).

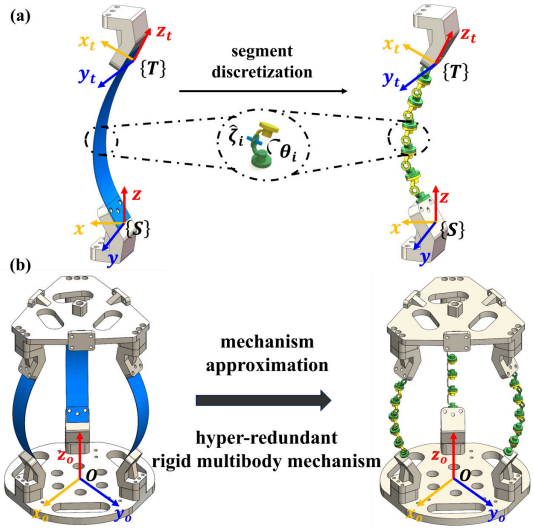


Fig. 3. Mechanism approximation of the slender elastic beam and the proposed end-effector. (a) Mechanism equivalent of the elasitc beam, (b) Mechanism approximation of the proposed end-effector

As depicted in the Fig. 3(a), the spatial frame and the tip frame of the elastic beam are denoted by {S} and {T}, respectively. The initial and final angles, corresponding to the segments where the beam's proximal and distal segments are attached to their connectors, are denoted by θ_A and θ_B, respectively, as illustrated in Fig. 2(c). Additionally, let $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_n]^T \in \mathbb{R}^{n\times 1}$ represent the vector of joint variables. The kinematics constraint at the beam's tip frame can be derived using the forward kinematcis formulated through the POE formula [26], expressed as follows:

$$\mathbf{g}_{st}(\boldsymbol{\theta}) = \mathbf{g}_{st}(A)\left(\exp\left(\hat{\boldsymbol{\zeta}}_1\theta_1\right)\cdots\exp\left(\hat{\boldsymbol{\zeta}}_n\theta_n\right)\mathbf{g}_{st,0}\right)\mathbf{g}_{st}(B) \quad (1)$$

where $\mathbf{g}_{st} \in \mathrm{SE}(3)$ defines the relative pose of $\{T\}$ with respect to $\{S\}$. $\mathbf{g}_{st,0}$ is represented as $\exp(\hat{\boldsymbol{\zeta}}_1\theta_1)\cdots\exp(\hat{\boldsymbol{\zeta}}_n\theta_n)$, with all θ_i set to zero.

Therefore, the initial pose is $\mathbf{g}_{st}(0) = \mathbf{g}_{st}(A)\mathbf{g}_{st,0}\mathbf{g}_{st}(B)$, where $\mathbf{g}_{st}(A) = \exp\left(\hat{\boldsymbol{\zeta}}_A\theta_A\right)$ and $\mathbf{g}_{st}(B) = \exp\left(\hat{\boldsymbol{\zeta}}_B\theta_B\right)$. $\hat{\boldsymbol{\zeta}}_A$ and $\hat{\boldsymbol{\zeta}}_B$ are the joint twists at the fixed connections at the start and end of the flexible beam to the connectors, respectively, while $\hat{\boldsymbol{\zeta}}_i$ for $i = 1,\ldots,n$ are the joint twists of the equivalent mechanism of the elastic beam. All these joint twists are elements of $se(3)$ and are represented in 4×4 standard form.

In the joint space, the restoring torques from the elastic joints are counterbalanced by the reaction forces resulting from the external load applied at the tip. The equilibrium state is formulated in the equation below:

$$\boldsymbol{\tau} = \boldsymbol{K}_\theta\boldsymbol{\theta} - \boldsymbol{J}^T\boldsymbol{F} = 0 \quad (2)$$

where $\boldsymbol{K}_\theta = diag(k_1,\ldots,k_n) \in \mathbb{R}^{n\times n}$ defines the diagonal stiffness matrix of the elastic joints, influenced by the mechanical characteristics of the beam. The external wrench exerted on the tip frame at $\{T\}$ is represented by $\boldsymbol{F} \in \mathbb{R}^{6\times 1}$. Since the deformation process occurs only within x-y plane, the wrench includes only three non-zero values, represented by $\boldsymbol{F} = [0,0,M_z,f_x,f_y,0]^T$. Additionally, $\boldsymbol{J} \in \mathbb{R}^{6\times n}$ denotes the Jacobian matrix that maps joint space movements to the tip frame, derived as follows:

$$\left(\dot{\mathbf{g}}_{st}\mathbf{g}_{st}^{-1}\right)^{\vee} = \boldsymbol{J}\dot{\boldsymbol{\theta}} \Rightarrow \boldsymbol{J} = [\boldsymbol{\xi}_1,\boldsymbol{\xi}_2,\ldots,\boldsymbol{\xi}_n] \tag{3}$$

Here, $(\cdot)^{\vee}$ denoted the 'vee' operator, which transforms elements from the Lie algebra $se(3)$, represented as 4×4 matrices, into 6×1 vectors. The vector $\boldsymbol{\xi}_i \in \mathbb{R}^{6\times 1}$ corresponds to the joint twist of the mechanism under its current configuration, described by $\boldsymbol{\theta}$. Utilizing the POE formula, $\boldsymbol{\xi}_i$ us derived as follows:

$$\boldsymbol{\xi}_i = \mathrm{Ad}\left(\exp(\hat{\boldsymbol{\zeta}}_A\theta_A)\exp(\hat{\boldsymbol{\zeta}}_1\theta_1)\cdots\exp(\hat{\boldsymbol{\zeta}}_{i-1}\theta_{i-1})\right)\boldsymbol{\zeta}_i \tag{4}$$

where $\mathrm{Ad}(\cdot)$ represents the adjoint operator for $SE(3)$. For further details, please refer to [27]. The synthesis of geometric constrains (4) at the tip with the static equilibrium conditions (5) within the joint space allows for the formulation of a kinetostatic model of the equivalent mechanism. This model is formulated as a set of nonlinear algebraic equations, specified as follows:

$$f_1(\boldsymbol{\theta},\boldsymbol{F}) = \begin{bmatrix}\Delta\boldsymbol{X}\\ \Delta\boldsymbol{\tau}\end{bmatrix} = \begin{bmatrix}\left(\ln(\mathbf{g}_{st}\mathbf{g}_t^{-1})\right)^V\\ \boldsymbol{K}_\theta\boldsymbol{\theta} - \boldsymbol{J}^T\boldsymbol{F}\end{bmatrix} = \mathbf{0} \tag{5}$$

Here, $\Delta\boldsymbol{X} \in \mathbb{R}^{6\times 1}$ denotes the pose deviation of the beam tip frame relative to the target frame $\mathbf{g}_t$, and $\boldsymbol{\tau} \in \mathbb{R}^{n\times 1}$ denotes the deviations in torques at the elastic joints, which are necessary to counterbalance the external load applied at the tip.

Then, the kinetostatics model for the end-effector is represented by an extended set of nonlinear algebraic equations, which are specified as follows:

$$f_2(\boldsymbol{\theta}_t,\boldsymbol{F}_t) = \begin{bmatrix}\Delta\boldsymbol{X}_1\\ \Delta\boldsymbol{X}_2\\ \Delta\boldsymbol{F}\\ \Delta\boldsymbol{\tau}_1\\ \Delta\boldsymbol{\tau}_2\\ \Delta\boldsymbol{\tau}_3\end{bmatrix} = \begin{bmatrix}\left(\ln(\mathbf{g}_{st,1}\mathbf{g}_{st,2}^{-1})\right)^V\\ \left(\ln(\mathbf{g}_{st,2}\mathbf{g}_{st,3}^{-1})\right)^V\\ \boldsymbol{F}_e - \boldsymbol{F}_1 - \boldsymbol{F}_2 - \boldsymbol{F}_3\\ \boldsymbol{K}_{\theta,1}\boldsymbol{\theta}_1 - \boldsymbol{J}_1^T\boldsymbol{F}_1\\ \boldsymbol{K}_{\theta,2}\boldsymbol{\theta}_2 - \boldsymbol{J}_2^T\boldsymbol{F}_2\\ \boldsymbol{K}_{\theta,3}\boldsymbol{\theta}_3 - \boldsymbol{J}_3^T\boldsymbol{F}_3\end{bmatrix} = 0 \tag{6}$$

where $\boldsymbol{\theta}_t = [\boldsymbol{\theta}_1,\boldsymbol{\theta}_2,\boldsymbol{\theta}_3]^T \in \mathbb{R}^{3n\times 1}$ denotes the equivalent joint values for each of the three beams. The wrench vector $\boldsymbol{F}_t = [\boldsymbol{F}_1,\boldsymbol{F}_2,\boldsymbol{F}_3]^T \in \mathbb{R}^{18\times 1}$ characterizes the interaction forces and torques between each beam i and the end-effector. Furthermore, $\mathbf{g}_{i,st} \in SE(3)$ represents the final stable pose of each beam. Under equilibrium conditions, both the geometric deviation $\Delta\boldsymbol{X}_i \in \mathbb{R}^{6\times 1}$ and the wrench deviation $\Delta\boldsymbol{F} \in \mathbb{R}^{6\times 1}$, as well as the joint force/torque offset $\boldsymbol{\tau}_i \in \mathbb{R}^{n\times 1}$, are expected to converge to zero, indicating system stability and balance.

B. Kinetostatics analysis

The kinetostatics model of the proposed end-effector consists of a set of $3n+18$ nonlinear algebraic equations, including the joint values $\boldsymbol{\theta}_t \in \mathbb{R}^{3n\times 1}$ and the force vector $\boldsymbol{F}_t \in \mathbb{R}^{18\times 1}$ as the unknowns. These components ensure the system is balanced and theoretically solvable with exact solutions. Subsequently, the derivation of associated gradient equations is outlined as follow:

$$\nabla_{f_2} = \begin{bmatrix}\frac{\partial\Delta\boldsymbol{X}_1}{\partial\boldsymbol{\theta}_t} & \frac{\partial\Delta\boldsymbol{X}_2}{\partial\boldsymbol{\theta}_t} & \frac{\partial\Delta\boldsymbol{F}}{\partial\boldsymbol{\theta}_t} & \frac{\partial\boldsymbol{\tau}_1}{\partial\boldsymbol{\theta}_t} & \frac{\partial\boldsymbol{\tau}_2}{\partial\boldsymbol{\theta}_t} & \frac{\partial\boldsymbol{\tau}_3}{\partial\boldsymbol{\theta}_t}\\ \frac{\partial\Delta\boldsymbol{X}_1}{\partial\boldsymbol{F}_t} & \frac{\partial\Delta\boldsymbol{X}_2}{\partial\boldsymbol{F}_t} & \frac{\partial\Delta\boldsymbol{F}}{\partial\boldsymbol{F}_t} & \frac{\partial\boldsymbol{\tau}_1}{\partial\boldsymbol{F}_t} & \frac{\partial\boldsymbol{\tau}_2}{\partial\boldsymbol{F}_t} & \frac{\partial\boldsymbol{\tau}_3}{\partial\boldsymbol{F}_t}\end{bmatrix}^T \tag{7}$$

The partial differential components of the gradient are computed as

$$\begin{bmatrix}\frac{\partial\Delta\boldsymbol{X}_1}{\partial\boldsymbol{\theta}_t} & \frac{\partial\Delta\boldsymbol{X}_2}{\partial\boldsymbol{\theta}_t}\end{bmatrix}^T = \begin{bmatrix}\boldsymbol{J}_1 & -Ad(\mathbf{g}_{1,st}\mathbf{g}_{2,st}^{-1})\boldsymbol{J}_2 & \mathbf{0}\\ \mathbf{0} & \boldsymbol{J}_2 & -Ad(\mathbf{g}_{2,st}\mathbf{g}_{3,st}^{-1})\boldsymbol{J}_3\end{bmatrix} \tag{8}$$

$$\left[\frac{\partial\Delta\boldsymbol{F}}{\partial\boldsymbol{\theta}_t}\right]^T = \begin{bmatrix}\mathbf{0} & \mathbf{0} & \mathbf{0}\end{bmatrix} \tag{9}$$

$$\begin{bmatrix}\frac{\partial\Delta\boldsymbol{\tau}_1}{\partial\boldsymbol{\theta}_t} & \frac{\partial\Delta\boldsymbol{\tau}_2}{\partial\boldsymbol{\theta}_t} & \frac{\partial\Delta\boldsymbol{\tau}_3}{\partial\boldsymbol{\theta}_t}\end{bmatrix}^T = \begin{bmatrix}\boldsymbol{K}_{\theta,1}-\boldsymbol{K}_{J,1} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{K}_{\theta,2}-\boldsymbol{K}_{J,2} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \boldsymbol{K}_{\theta,3}-\boldsymbol{K}_{J,3}\end{bmatrix} \tag{10}$$

$$\begin{bmatrix}\frac{\partial\Delta\boldsymbol{X}_1}{\partial\boldsymbol{F}_t} & \frac{\partial\Delta\boldsymbol{X}_2}{\partial\boldsymbol{F}_t}\end{bmatrix}^T = \begin{bmatrix}\mathbf{0}\\ \mathbf{0}\end{bmatrix} \tag{11}$$

$$\left[\frac{\partial\Delta\boldsymbol{F}}{\partial\boldsymbol{F}_t}\right]^T = \begin{bmatrix}\boldsymbol{I}_{6\times 6} & \boldsymbol{I}_{6\times 6} & \boldsymbol{I}_{6\times 6}\end{bmatrix} \tag{12}$$

$$\begin{bmatrix}\frac{\partial\Delta\boldsymbol{\tau}_1}{\partial\boldsymbol{F}_t} & \frac{\partial\Delta\boldsymbol{\tau}_2}{\partial\boldsymbol{F}_t} & \frac{\partial\Delta\boldsymbol{\tau}_3}{\partial\boldsymbol{F}_t}\end{bmatrix}^T = \begin{bmatrix}\boldsymbol{J}_1^T & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{J}_2^T & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \boldsymbol{J}_3^T\end{bmatrix} \tag{13}$$

where $\boldsymbol{J}_i$, for $i = 1,2,3$, denotes the Jacobian matrix associated with beam i. $\boldsymbol{K}_{\theta,i}$, for $i = 1,2,3$, represents the stiffness matrix for beam i, and $\boldsymbol{K}_{J,i}$, for $i = 1,2,3$, corresponds to the configuration-dependent item, derived as Ref.[28].

Subsequently, gradient-based search algorithms is applicable for addressing the kinematic problem associated with the developed end-effector, expressed as follows

$$\boldsymbol{X}_{f_2}^{(k+1)} = \boldsymbol{X}_{f_2}^{(k)} - (\nabla_{f_2}^{(k)})^{-1} f_2^{(k)} \tag{14}$$

where $\boldsymbol{X}_{f_2} = [\boldsymbol{\theta}_t,\boldsymbol{F}_t]^T \in \mathbb{R}^{(3n+18)\times 1}$ denotes the vector of unknown variables to be determined. The derivation and solution process outlined can be programmed using common mathematical software to efficiently compute the results.

IV. Sensing Method and Modeling

As indicated in Section II, flexible strain gauges are employed to measure the deflections of the elastic beams. On the basis of the kinetostatic model, the constraints on local deflections can be employed to reconstruct the deformation process of the elastic beam. Simultaneously, the corresponding output forces can be determined from the sensing model. Therefore, the developed end-effector is capable of achieving contact force control based on sensory feedback.

A. Sensing model of the elastic beam

Take the modeling of slender elastic beam elaborated in Section III as an instance, one strain gauge is deployed on this elastic beam, from which the deflections of the corresponding locations can be measured directly, as shown in Fig. 4. Then, extra constraints, in addition to those in (5), can be obtained according to the local deflection as

$$f_3(\boldsymbol{\theta},\boldsymbol{F},\boldsymbol{h}) = \begin{bmatrix} \Delta\boldsymbol{X} \\ \Delta\boldsymbol{\tau} \\ \Delta\mathbf{s} \end{bmatrix} = \begin{bmatrix} \left(\ln(\mathbf{g}_{st}\mathbf{g}_t^{-1})\right)^V \\ \boldsymbol{K}_\theta\boldsymbol{\theta} - \boldsymbol{J}^T\boldsymbol{F} \\ \mathbf{S_e}\boldsymbol{\theta} - s_0 \end{bmatrix} = \mathbf{0} \tag{15}$$

where $\boldsymbol{h} \in \mathbb{R}^{1\times 1}$ represents the displacement of the elastic beam due to deformation within the plane, and $s_0 \in \mathbb{R}^{1\times 1}$ represents the bending angle reading from the deflection sensors. While $\mathbf{S}_e \in \mathbb{R}^{1\times n}$ represents the selection matrix that correlates with the placement positions of the strain gauges on the beam. It is worth noting that the change of resistance in the employed strain gauges are linearly proportional to the local bending of the beam. For further details, see references [29], [30] and [31].

For the established sensing model, the number of unknowns increases to $n+7$, and the addition of equation $\boldsymbol{s} \in \mathbb{R}^{1\times 1}$ ensures that the number of equations matches the number of unknowns. Similar to the solution process in Section III, the gradient equations can be derived as follow:

$$\nabla_{f_3} = \begin{bmatrix} \frac{\partial\Delta\boldsymbol{X}}{\partial\boldsymbol{\theta}} & \frac{\partial\Delta\boldsymbol{X}}{\partial\boldsymbol{F}} & \frac{\partial\Delta\boldsymbol{X}}{\partial\boldsymbol{h}} \\ \frac{\partial\boldsymbol{\tau}}{\partial\boldsymbol{\theta}} & \frac{\partial\boldsymbol{\tau}}{\partial\boldsymbol{F}} & \frac{\partial\boldsymbol{\tau}}{\partial\boldsymbol{h}} \\ \frac{\partial\boldsymbol{s}}{\partial\boldsymbol{\theta}} & \frac{\partial\boldsymbol{s}}{\partial\boldsymbol{F}} & \frac{\partial\boldsymbol{s}}{\partial\boldsymbol{h}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{J} & \mathbf{0} & \boldsymbol{p} \\ \boldsymbol{K}_\theta - \boldsymbol{K}_J & -\boldsymbol{J}^T & \mathbf{0} \\ \boldsymbol{S}_e & \mathbf{0} & 0 \end{bmatrix} \tag{16}$$

where $\boldsymbol{p} = [0,0,0,-1,0,0]^T$. Once the gradients of the sensing model are determined, we still employ iterative algorithm to solve this problem. This approach not only facilitates the reconstruction of the deformation state but also enables the calculation of the output force $\boldsymbol{F}$ in the current configuration.

B. Calibration of the deformed beam

To establish a relationship between the voltage readings of the strain gauge and the local deformation angles of the elastic beam, a calibration process is essential. As illustrated in Fig. 4, a single-DOF measurement platform is developed. The upper and lower ends of the elastic beam are connected to a slider and a base, respectively. Equipped with a laser sensor, the measurement of the distance between the two ends of the beam is facilitated. These measurements are then incorporated into the kinetostatic model established in Section III-A to calculate the local deformation angles of the beam. According to the characteristics of the flexible strain gauge, a linear relation between the deflection curvatures c and the voltage $\mathbf{U}_s$ of the strain gauge can be expressed as

$$c = \frac{2l_0}{GFh_0U_0}\mathbf{U}_s = \lambda\mathbf{U}_s \tag{17}$$

where h_0 relates to the thickness of beam and l_0 to the effective length of the gauges. GF denotes the sensitivity ratio, and U_0 represents the supply voltage of the strain gauge. Ultimately, the above parameters can be consolidated into a single linear parameter λ , which can be determined through calibration in practical applications.

Within the permissible deformation range, 22 configurations of the elastic beam are selected for measurement. As shown in Fig. 4, the relationship between the recorded voltage outputs of the strain gauges and the local bending curvatures calculated through the beam's kinetostatics is displayed. These observations confirm a strong linear relationship, allowing for the establishment of linear formulas for each strain sensor's curvature-voltage relationship.

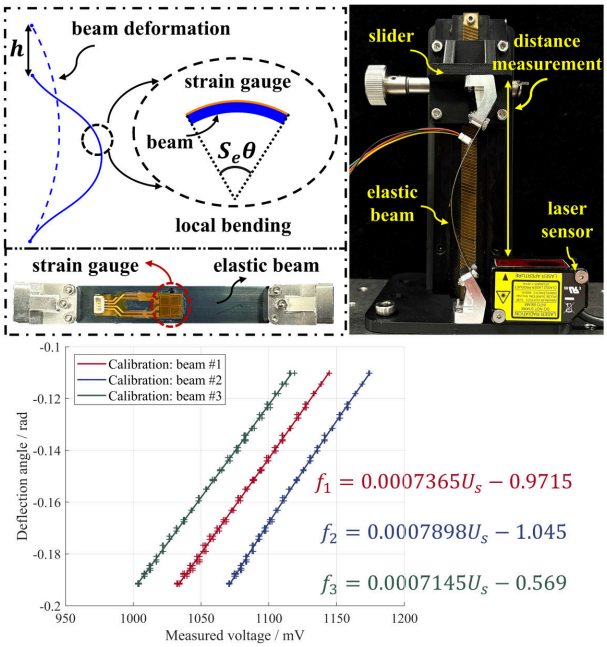


Fig. 4. Deflection sensing method using strain gauges: theory and calibration

V. Experimental Validation

In this section, a variety of experiments are conducted on the developed prototype to validate the feasibility of the proposed end-effector, and assess the effectiveness of the established kinetostatics model and sensing methods. From the results, it is known that the developed force-controlled end-effector is able to control output forces with the model-based feedback control. Consequently, the end-effector can effectively accomplish robotic grinding task.

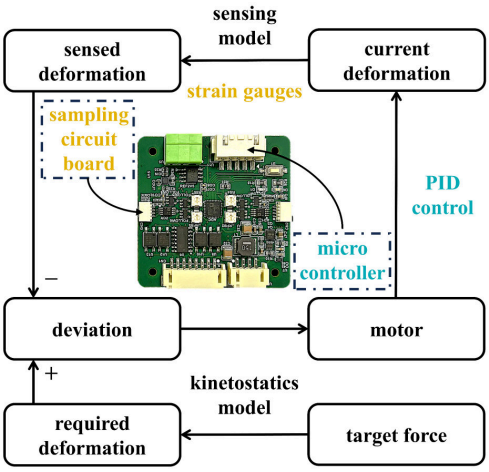


Fig. 5. Control strategy of the proposed end-effector implemented through an integrated control and sampling module

A. Control strategy of the end-effector

In Section II-B, the materials and fabrication methods of the developed end-effector are described. The control strategy of the end-effector is detailed through a block diagram presented in Fig. 5. This block diagram includes a kinetostatics model that calculates the required deformation state of the elastic beam based on a target contact force. Additionally, a sensing model based on strain gauges is employed to perceive the current deformation state of the beam. Deviations between the required and sensed deformation states are used to control the motor, thus achieving a new deformation state for the elastic beam. The operation of the system is facilitated by a specially developed, highly integrated module. This module incorporates an STM32 micro controller, which is tasked with motor control. It also includes a sampling circuit that is crucial for the collection and processing of signals from the strain gauges. Moreover, considering the diverse requirements of different testing scenarios, the end-effector is mounted on the end of a UR10 robotic arm. This robotic arm is controlled by a C++ based GUI software that enables real-time monitoring and adjustment of the robotic arm's Tool Center Point (TCP) position and joint angles.

B. Stability of the end-effector output force

As indicated in the above, the elastic beams constitute the force-generating unit of the proposed end-effector. Consequently, evaluating the stability of the end-effector's output forces is critical. An experimental setup, as illustrated in Fig. 6(a), employs a six-axis force sensor (SRI-M3813B) to measure these output forces. Within the permissible driving range, 30 different actuation values are selected for testing. Each configuration is subjected to five measurements of the output force to ensure reliability of the data.

Fig. 6(b) illustrates the maximum and minimum contact forces measured under various driving values, alongside the observed fluctuations in output forces. Within the allowable range of motor operation, the proposed end-effector can generate a maximum output force of approximately $37N$. The fluctuation is determined by calculating the difference between the maximum and minimum force values recorded in each experimental set. This fluctuation in output force does not exceed $0.55N$. Such results demonstrate that the slender elastic beam, integral to the force-generating unit of the developed end-effector, are capable of maintain a stable force output.

C. Validation of kinetostatics model and sensing method

In this section, experiments are conducted to evaluate the effectiveness of the proposed end-effector's kinetostatic model in controlling output forces and to validate the developed sensing method. The kinetostatic model is utilized to calculate the required driving values for specified contact forces, and the actual measured forces are compared with the designed forces. Concurrently, the sensed forces are calculated based on voltage values from strain gauges using the sensing method, and these sensed forces are compared with the actual measured forces to assess effectiveness of the sensing methodology. Furthermore, a laser sensor, as shown in Fig. 7(a), is integrated into the experimental setup to ensure that the calculated driving values are accurately executed on the end-effector.

As depicted in Fig. 7(b), the fluctuations in the measured output forces exhibit trends similar to those shown in Fig. 6(b), further demonstrating the stability of the developed end-effector. The deviation between the measured output forces and the designed output forces is represented by the blue curve

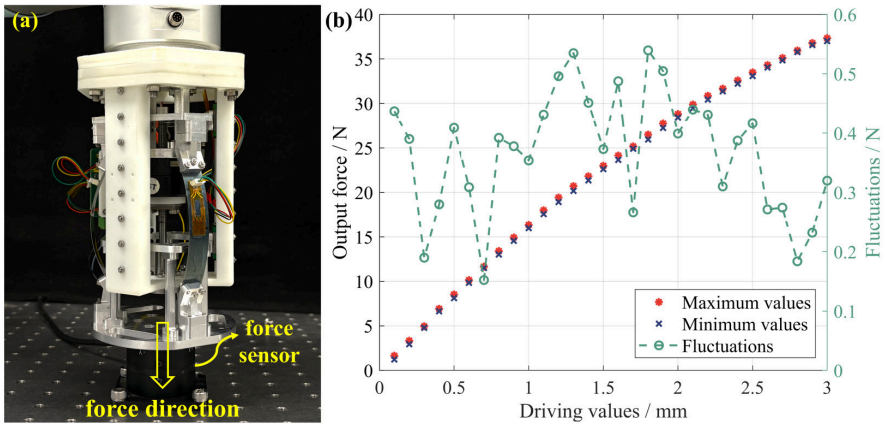


Fig. 6. Evaluation of the output force stability. (a) Experimental setup, (b) Fluctuations in output force

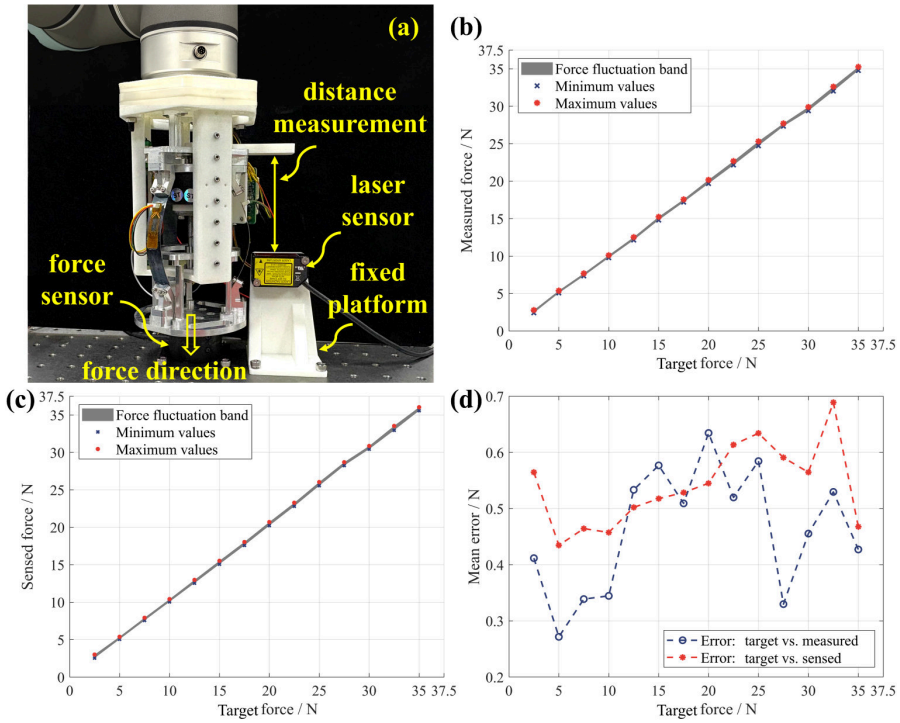


Fig. 7. Validation of kinetostatic model and sensing method. (a) Experimental setup, (b) Measured contact forces, (c) Sensed contact forces, (d) Comparison of measured and designed forces, sensed and designed forces

in Fig. 7(d). The maximum average error in contact forces does not exceed 0.65N. This indicates that the established kinetostatic model is capable of calculating the motor drive required to achieve the target contact force.

The sensing results of the output force, as shown in Fig. 7(c), demonstrate that the voltage readings from the strain gauge are consistent and can reliably measure local deformation. Additionally, the red curve in Fig. 7(d) displays the error between the sensed force, calculated via the sensing model, and the designed contact force. Across 14 different output force experiments, the average error remains below 0.7N, indicating that the established sensing model can utilize the voltage readings from the strain gauge to calculate the output force. Therefore, the developed end-effector is capable of sensing the output force using the strain gauge, which lays a solid foundation for controlling contact force based on sensory feedback.

To thoroughly validate the effectiveness of the established model, experiments incorporating variable contact force control are performed. As illustrated in Fig. 8 and 9, the control of contact forces following sinusoidal and stepped trajectories is executed individually. Utilizing the model, the necessary motor drive calculations are made to ensure that the end-effector can produce the desired contact force trajectories. The RMS error of force control in the two prescribed trajectories are 0.44N and 0.32N respectively. Additionally, the maximum errors observed are 1N for the sinusoidal trajectory and 0.75N for the stepped trajectory. Notably, the increase in error predominantly occur at the peaks and troughs of the contact force curves, which can be attributed to inertia effects during the motion process.

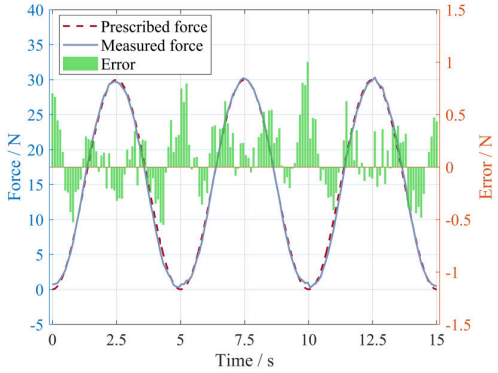


Fig. 8. Variable contact force control (Sinusoidal contact force)

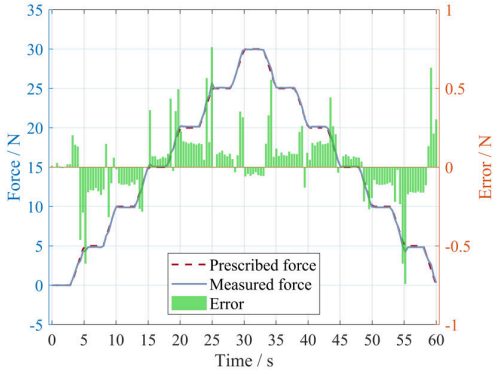


Fig. 9. Variable contact force control. (Stepped contact force)

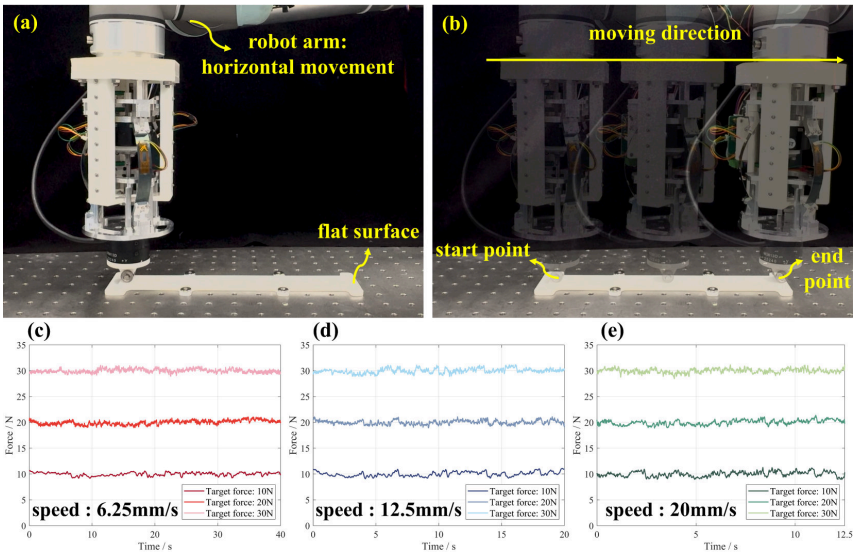


Fig. 10. Effectiveness of contact force control based on sensory feedback (flat surface on the horizontal scene). (a) Experimental setups, (b) Experimental process, (c-e) Contact force data at different robotic arm speeds

D. Constant contact force control based on self-sensing

During the grinding process, the surface of the workpiece is usually uneven, and there are also deviations in the movement of the robotic arm. Consequently, fully pre-programmed control of the end-effector presents considerable challenges. As addressed in Section V-C, the end-effector, equipped with strain gauges attached to elastic beams, is capable of sensing the output force based on the sensing model. After calculating the sensed forces, the end-effector can autonomously control contact forces based on sensory feedback. A set of pulleys is installed at the output end of the end-effector, then the robotic arm is controlled to execute linear movements in different scenarios. Based on sensory feedback, this end-effector can maintain a constant contact force throughout the motion process.

The design of the first contact surface features a 250mm long flat plane, as shown in Fig. 10(a). To evaluate the control capabilities of the developed end-effector, constant contact forces of 10N, 20N, and 30N are set as control targets. The robotic arm is programmed to move horizontally at varying speeds to test the responsiveness and precision of the end-effector. Fig. 10(c-e) illustrate the end-effector's ability to maintain these specified contact forces at different speeds of the robotic arm's movement.

As shown in the Fig. 10(c-e), the proposed end-effector demonstrates the capability to maintain a specified output force within a certain fluctuation range during the movement of the robotic arm. Detailed data on these fluctuations is presented in Table II, where the fluctuation is quantified as the difference between the maximum and minimum contact forces relative to the required contact force. When the robotic arm moves at speeds of $5mm/s$ and $12.5mm/s$, the fluctuation in the $10N$ contact force remains within $\pm 1N$. At a higher speed of $20mm/s$ and with a controlled contact force of $30N$, the fluctuation increases to $\pm 1.35N$.

The design of the second contact surface is shaped as a sinusoidal curve, as illustrated in Fig. 11(a), to simulate undulations of a workpiece surface. The settings for the specified contact force and the speed of the robotic arm, along with the experimental apparatus, are consistent with those used in the experiments with the first contact surface. Detailed results are displayed in Fig. 11(c-e) and Table III.

TABLE II
Fluctuation results of the controlled contact force
(flat surface on the horizontal scene)

	Speed (mm/s)		
Force (N)	6.25	12.5	20
10	$\pm 0.74N$	$\pm 0.99N$	$\pm 1.16N$
20	$\pm 1.02N$	$\pm 1.06N$	$\pm 1.24N$
30	$\pm 1.05N$	$\pm 1.13N$	$\pm 1.35N$

TABLE III
Fluctuation results of the controlled contact force
(curved surface on the horizontal scene)

	Speed (mm/s)		
Force (N)	6.25	12.5	20
10	$\pm 1.55N$	$\pm 1.54N$	$\pm 1.78N$
20	$\pm 1.68N$	$\pm 1.73N$	$\pm 1.80N$
30	$\pm 1.88N$	$\pm 2.04N$	$\pm 2.05N$

The results indicate that the fluctuations in the contact force follow a sinusoidal pattern, mirroring the design of the curved contact surface. On the curved surface, when the required contact force is set below $30N$, the fluctuations remain below $1.9N$ across various speeds of the robotic arm, as detailed in Table III. When the contact force is increased to $30N$, the fluctuation reaches $2N$ as the speed of the robotic arm increases. Despite the fact that higher speeds and greater forces lead to increased fluctuations, these increments are relatively small.

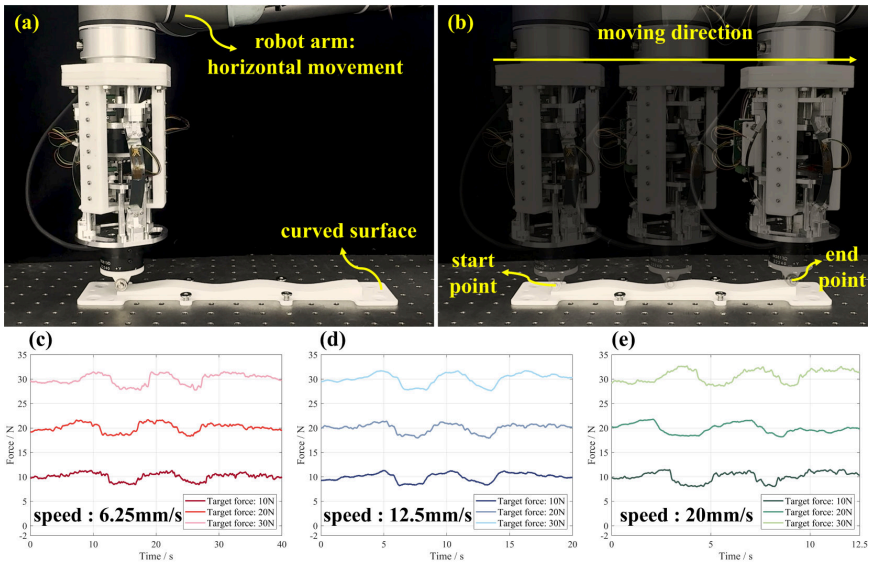


Fig. 11. Effectiveness of contact force control based on sensory feedback (curved surface on the horizontal scene). (a) Experimental setups, (b) Experimental process, (c-e) Contact force data at different robotic arm speeds

In order to more comprehensively evaluate the contact force control performance of the proposed end-effector in different environments, experimental setups for both sloped and vertical scenarios are designed. The designed sinusoidal contact surface is fixed on a slope with a 30-degree incline. The motion trajectory of the robotic arm is set parallel to the slope surface, as illustrate in Fig. 12. The specified contact force and the movement speed of the robotic arm are consistent with the experimental setup in the horizontal scenario. Detailed results of these tests are presented in Fig. 12(c-e) and Table. IV.

The results demonstrate that the proposed design achieves comparable performance in sloped environments as seen in horizontal settings. When the robotic arm operates at a speed of $6.25mm/s$, fluctuations in contact force of various magnitude does not exceed $\pm 1.8N$. Even at the highest moving speeds of the robotic arm and with a controlled contact force of $30N$, the fluctuation of the contact force remains within $\pm 2N$.

TABLE IV
Fluctuation results of the controlled contact force
(curved surface on the slope scene)

	Speed (mm/s)		
Force (N)	6.25	12.5	20
10	$\pm 1.52N$	$\pm 1.69N$	$\pm 1.78N$
20	$\pm 1.61N$	$\pm 1.71N$	$\pm 1.84N$
30	$\pm 1.79N$	$\pm 1.97N$	$\pm 1.98N$

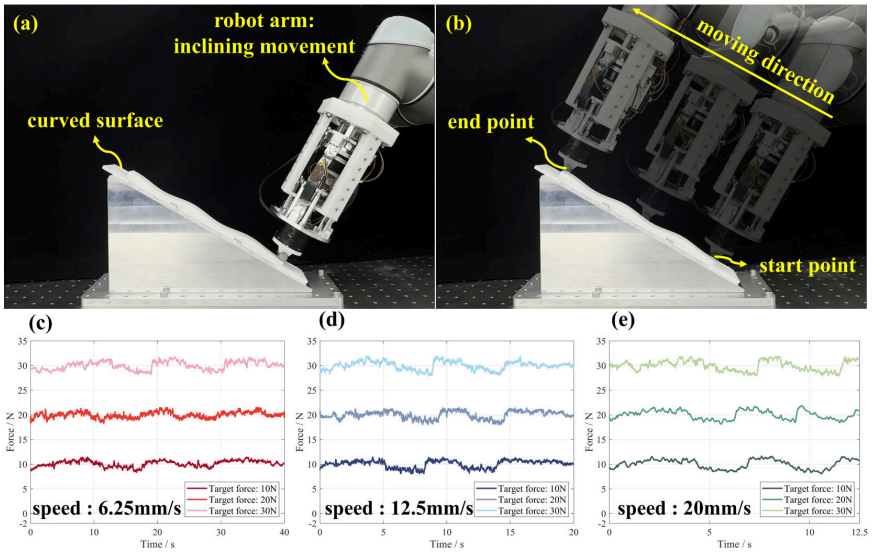


Fig. 12. Effectiveness of contact force control based on sensory feedback (curved surface on the slope scene). (a) Experimental setups, (b) Experimental process, (c-e) Contact force data at different robotic arm speeds

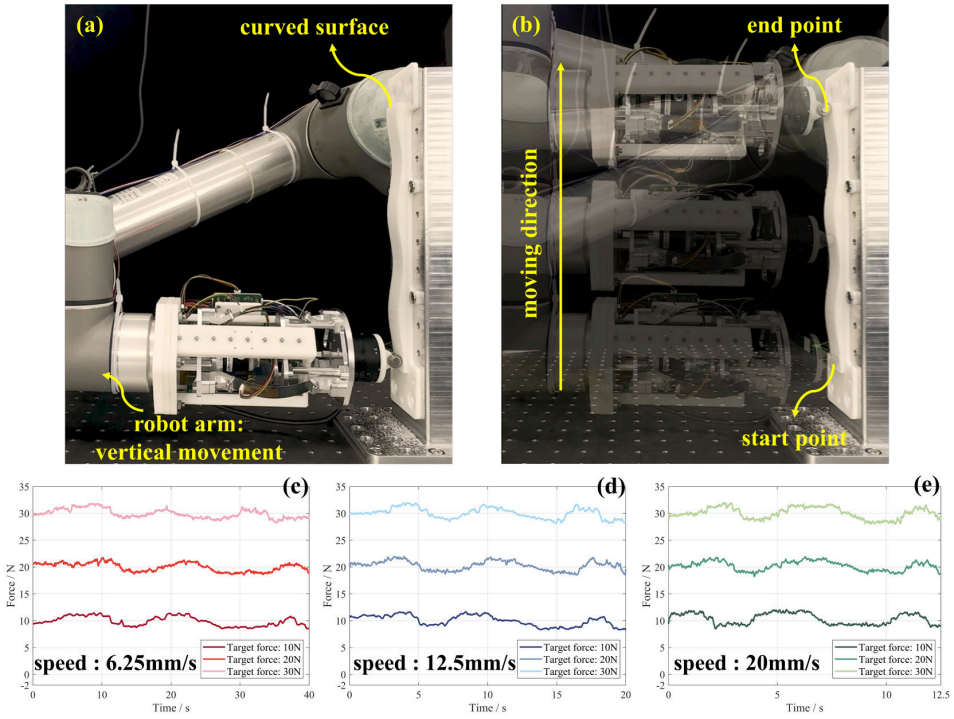


Fig. 13. Effectiveness of contact force control based on sensory feedback (curved surface on the vertical scene). (a) Experimental setups, (b) Experimental process, (c-e) Contact force data at different robotic arm speeds

Tests on a curved surface in a vertical configuration, which involved the robotic arm moving vertically, are also conducted as shown in Fig. 13(b). The results, as depicted in Fig. 13(c-e) and Tab. V, demonstrate that the end-effector maintains similar control performance over contact forces in vertical scene as it does in horizontal and slope scenarios. The results indicate that the proposed force-controlled end-effector can manage contact forces in both inclined and vertical scenarios.

TABLE V
Fluctuation results of the controlled contact force
(curved surface on the vertical scene)

	Speed (mm/s)		
Force (N)	6.25	12.5	20
10	$\pm 1.47N$	$\pm 1.68N$	$\pm 1.79N$
20	$\pm 1.59N$	$\pm 1.70N$	$\pm 1.82N$
30	$\pm 1.81N$	$\pm 1.93N$	$\pm 1.99N$

As the robotic arm's speed increases, the proposed force-controlled end-effector can boost responsiveness by raising the strain gauge sampling rate and motor control frequency. However, it becomes more susceptible to output force overshoot and instability. To ensure stable and repeatable control, the current three moving speeds are used to evaluate the proposed end-effector. For operation on curved surfaces, small curvature approximates a flat surface and yields smaller contact force fluctuations. As curvature increases, the single-DOF force control capability becomes a limiting factor for force control performance, motivating future design optimization.

E. Demonstration of grinding application

To further evaluate the force control capabilities of the proposed end-effector during practical grinding operations, a grinding tool is attached to the end of the effector. Experiments are executed on a workpiece featuring a fully black oxidized surface, as shown in Fig. 14(a). Two sets of experiments are conducted with the robotic arm executing a horizontal motion: one with force control activated and the other without force control. Fig. 14(b) displays a comparison of the workpiece surface before and after grinding.

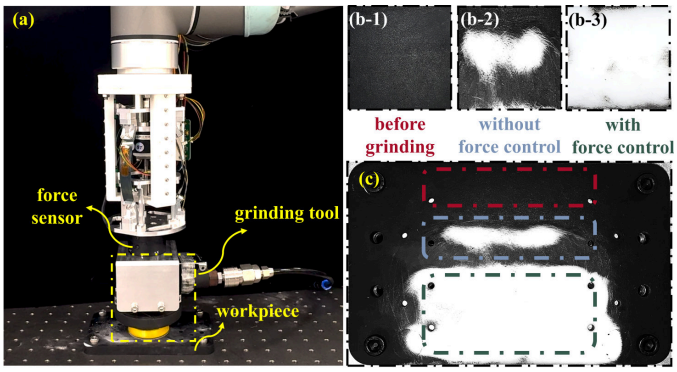


Fig. 14. Demonstration of the proposed end-effector's performance of robotic grinding (horizontal scenarios). (a) Experimental setup, (b-c) Contact forces during grinding: without and with force control

When force control is disabled, the black oxide layer is only partially removed, indicating inconsistent force application across the surface. Conversely, with force control enabled, the

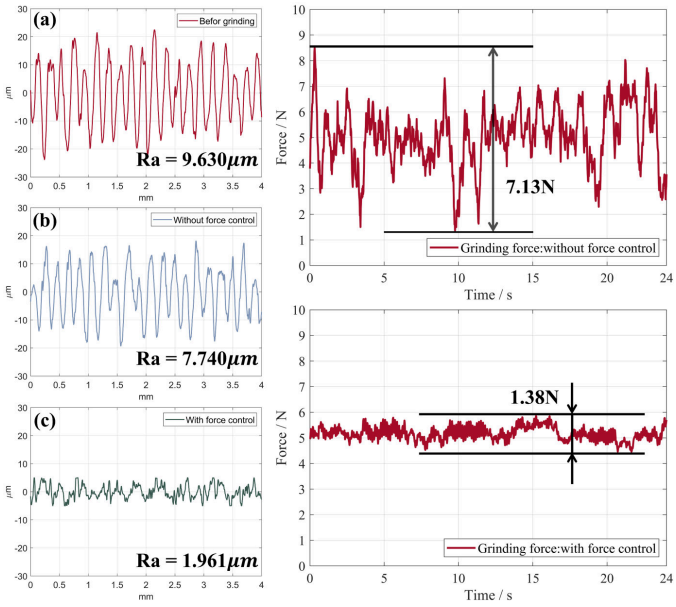


Fig. 15. Comparison of the surface roughness and contact forces during grinding. (a-c) Surface roughness results: before grinding, grinding without force control, grinding with force control, (d-e) Contact forces: without force control and with force control

black oxide layer is uniformly removed, demonstrating the effectiveness of active force control in achieving consistent grinding results. Additionally, the surface roughness of the workpiece before and after grinding is measured, as illustrated in Fig. 15(a-c). Initially, the roughness average (Ra) of the unground workpiece is 9.630 μm. After grinding with force sensing disabled, there is a reduction in roughness with the Ra decreasing to 7.740 μm. However, a more significant reduction in roughness is achieved with force sensing enabled, where the Ra dramatically decreases to 1.961 μm.

The results of contact force during the grinding process are

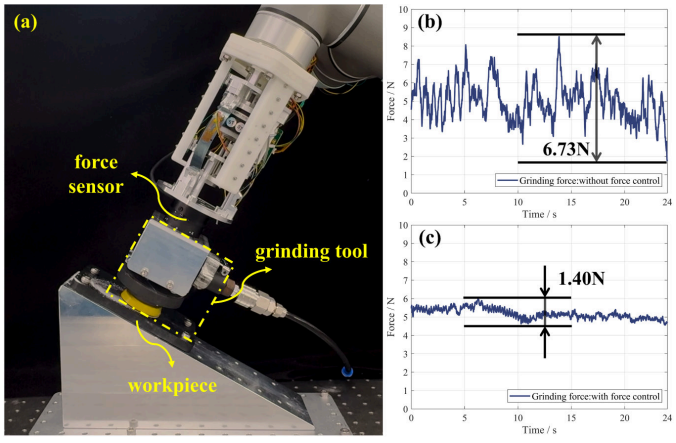


Fig. 16. Demonstration of the proposed end-effector's performance of robotic grinding (inclined scenarios). (a) Experimental setup, (b-c) Contact forces during grinding: without and with force control

illustrated in Fig. 15(d-e). The experiments are designed to maintain a constant contact force of 5N at the end-effector. In scenarios where force control is disabled, the contact force experiences significant fluctuations, reaching as high as 7.13N. Conversely, when force control is enabled, the fluctuation in contact force is significantly reduced to just 1.38N. Consequently, the developed end-effector demonstrates its capability to control contact force with sensory feedback, exhibiting the potential in achieving grinding tasks.

To comprehensively assess the grinding contact force control performance of the proposed end-effector across different orientations, grinding experiments are conducted in inclined and vertical scenarios. These experiments follow the methodology used in horizontal scenarios, where the robotic arm executes linear movements along the direction of the slope or vertically. Experiments with contact force control enabled and disabled are conducted, as illustrated in Fig.16 and 17.

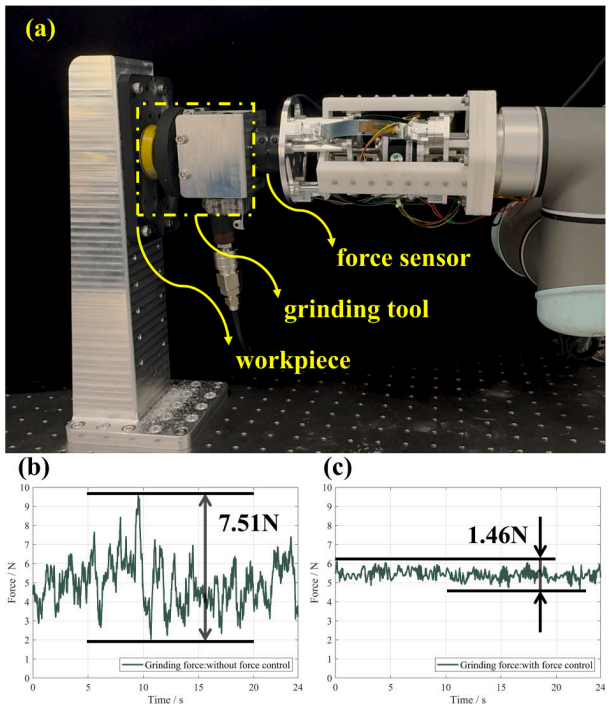


Fig. 17. Demonstration of the proposed end-effector's performance of robotic grinding (vertical scenarios). (a) Experimental setup, (b-c) Contact forces during grinding: without and with force control

The results observed from the Fig. 16(b-c) and 17(b-c) illustrate the effectiveness of the proposed end-effector with sensory-based feedback force control in reducing fluctuations in grinding contact forces. In the inclined scenarios, the fluctuation in contact force decreases from 6.73N to 1.40N. Similarly, in the vertical scenarios, the fluctuation is reduced from 7.51N to 1.46N. This performance in contact force control is similar with the results observed in horizontal scenario, demonstrating the end-effector's ability to adapt to various grinding environments. This suggests that the proposed design has the potential for grinding applications.

VI. Conclusion

In this paper, a force-controlled end-effector with the capabilities of passive structural compliance and intrinsic contact sensing is developed. By incorporating slender elastic beams, the proposed end-effector not only generates force but also provides passive compliance. Flexible strain gauges integrated into the elastic beam enable the end-effector to sense contact forces within environments. A closed-form kinetostatic model is established, which allows the end-effector to perceive contact forces through the local deflections of strain gauges.

A proof-of-concept prototype of the proposed end-effector is developed, on which a variety of experiments are carried out. The results demonstrate that, through embedded strain gauges, the developed end-effector can stably generate and sense the desired contact forces with a maximum error of less than $0.7N$, thus achieving effective contact force control based on sensory feedback. In a grinding experiment on a real workpiece, the studied end-effector demonstrates an 80% reduction in contact force fluctuations, highlighting its potential in application scenarios that demand force control. For future research, a three-DOF end-effector for hybrid orientation/force control will be developed to overcome the limitation of the current design, which is restricted to single-DOF contact force control. In addition, the sensitivity of intrinsic contact sensing and accuracy of close-loop actuation based on sensory feedback will be enhanced to improve response speed and force control performance, enabling applicability in more complex scenarios.

References

[1] H. Huang, Z. Gong, X. Chen, and L. Zhou, "Robotic grinding and polishing for turbine-vane overhaul," *Journal of materials processing technology*, vol. 127, no. 2, pp. 140–145, 2002.

[2] A. E. K. Mohammad and D. Wang, "Electrochemical mechanical polishing technology: recent developments and future research and industrial needs," *The International Journal of Advanced Manufacturing Technology*, vol. 86, pp. 1909–1924, 2016.

[3] E. Pitschke, M. Schinhaerl, R. Rascher, P. Sperber, L. Smith, R. Stamp, and M. Smith, "Simulation of a complex optical polishing process using a neural network," *Robotics and Computer-Integrated Manufacturing*, vol. 24, no. 1, pp. 32–37, 2008.

[4] F. Nagata, Y. Kusumoto, Y. Fujimoto, and K. Watanabe, "Robotic sanding system for new designed furniture with free-formed surface," *Robotics and Computer-Integrated Manufacturing*, vol. 23, no. 4, pp. 371–379, 2007.

[5] J.-H. Ahn, Y. Shen, H. Kim, H.-D. Jeong, and K.-K. Cho, "Development of a sensor information integrated expert system for optimizing die polishing," *Robotics and Computer-Integrated Manufacturing*, vol. 17, no. 4, pp. 269–276, 2001.

[6] A. Sharon, N. Hogan, and D. E. Hardt, "The macro/micro manipulator: An improved architecture for robot control," *Robotics and computer-integrated manufacturing*, vol. 10, no. 3, pp. 209–222, 1993.

[7] Y.-H. Chen and C.-C. Lan, "An adjustable constant-force mechanism for adaptive end-effector operations," 2012.

[8] J. Zhang, L. Zhao, L. Li, F. Ma, and G. Chen, "Design of passive constant-force end-effector for robotic polishing of optical reflective mirrors," *Chinese Journal of Mechanical Engineering*, vol. 35, no. 1, p. 141, 2022.

[9] Y. Wei and Q. Xu, "Design of a new passive end-effector based on constant-force mechanism for robotic polishing," *Robotics and Computer-Integrated Manufacturing*, vol. 74, p. 102278, 2022.

[10] Z. Wu, X. Wang, Z. Lyu, and Q. Xu, "Design of a novel passive polishing end-effector with adjustable constant force and wide operating angle," *IEEE/ASME Transactions on Mechatronics*, 2024.

[11] G. Bone and M. Elbestawi, "Active end effector control of a low precision robot in deburring," *Robotics and computer-integrated manufacturing*, vol. 8, no. 2, pp. 87–96, 1991.

[12] Z. Ma, A. N. Poo, M. H. Ang Jr, G.-S. Hong, and F. Huo, "Design, simulation and implementation of a 3-puu parallel mechanism for a macro/mini manipulator." in *FinE-R@ IROS*, 2015, pp. 42–47.

[13] Z. Ma, A.-N. Poo, M. H. Ang Jr, G.-S. Hong, and H.-H. See, "Design and control of an end-effector for industrial finishing applications," *Robotics and Computer-Integrated Manufacturing*, vol. 53, pp. 240–253, 2018.

[14] A. E. K. Mohammad, J. Hong, and D. Wang, "Design of a force-controlled end-effector with low-inertia effect for robotic polishing using macro-mini robot approach," *Robotics and Computer-Integrated Manufacturing*, vol. 49, 2018.

[15] F. Chen, H. Zhao, D. Li, L. Chen, C. Tan, and H. Ding, "Contact force control and vibration suppression in robotic polishing with a smart end effector," *Robotics and Computer-Integrated Manufacturing*, vol. 57, pp. 391–403, 2019.

[16] Q. Wang, W. Wang, X. Ding, and C. Yun, "A force control joint for robot–environment contact application," *Journal of Mechanisms and Robotics*, vol. 11, no. 3, p. 034502, 2019.

[17] Q. Wang, W. Wang, L. Zheng, and C. Yun, "Force control-based vibration suppression in robotic grinding of large thin-wall shells," *Robotics and Computer-Integrated Manufacturing*, vol. 67, p. 102031, 2021.

[18] X. Bao, S. Wang, R. Housden, J. Hajnal, and K. Rhode, "A constant-force end-effector with online force adjustment for robotic ultrasonography," *IEEE robotics and automation letters*, vol. 6, no. 2, pp. 2547–2554, 2021.

[19] Y. Qin, H. Wu, Z. Li, N. Sun, and L. Sun, "Design and analysis of a compliant end-effector for robotic polishing using flexible beams," in *Actuators*, vol. 11, no. 10. MDPI, 2022, p. 284.

[20] H. Chen, J. Yang, and H. Ding, "Robotic compliant grinding of curved parts based on a designed active force-controlled end-effector with optimized series elastic component," *Robotics and Computer-Integrated Manufacturing*, vol. 86, p. 102646, 2024.

[21] G. Chen, Z. Zhang, and H. Wang, "A general approach to the large deflection problems of spatial flexible rods using principal axes decomposition of compliance matrices," *Journal of Mechanisms and Robotics*, vol. 10, no. 3, p. 031012, 2018.

[22] G. Chen, H. Wang, Z. Lin, and X. Lai, "The principal axes decomposition of spatial stiffness matrices," *IEEE Transactions on Robotics*, vol. 31, no. 1, pp. 191–207, 2015.

[23] G. Chen, S. Tang, X. Duan, and H. Wang, "Design, modeling, and evaluation of parallel continuum robots: A survey," *Science China Technological Sciences*, vol. 67, no. 3, pp. 673–695, 2024.

[24] G. Chen, Y. Kang, Z. Liang, Z. Zhang, and H. Wang, "Kinetostatics modeling and analysis of parallel continuum manipulators," *Mechanism and Machine Theory*, vol. 163, p. 104380, 2021.

[25] G. Chen, Y. Long, S. Yao, S. Tang, J. Luo, H. Wang, Z. Zhang, and H. Jiang, "A non-electrical pneumatic hybrid oscillator for high-frequency multimodal robotic locomotion," *Nature Communications*, vol. 16, no. 1, p. 1449, 2025.

[26] K. M. Lynch and F. C. Park, *Modern robotics*. Cambridge University Press, 2017.

[27] Y. Long, X. Zhu, P. Shi, Q. Liu, Y. Wang, H. Wang, and G. Chen, "A high-performance elastic-soft hybrid pneumatic actuator with origami structure," *International Journal of Mechanical Sciences*, vol. 287, p. 109935, 2025.

[28] X. Duan, W. Yan, T. Yan, G. Chen, and H. Wang, "Kinetostatics modeling and analysis of a spherical parallel continuum manipulator," *Journal of Mechanisms and Robotics*, vol. 17, no. 1, 2025.

[29] J. Luo, Y. Xun, J. Yao, G. Chen, and H. Wang, "Sensor-based reconstruction of slender flexible beams undergoing large-scale deflection," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2022, pp. 6936–6943.

[30] G. Chen, S. Tang, S. Xu, T. Guan, Y. Xun, Z. Zhang, H. Wang, and Z. Lin, "Intrinsic contact sensing and object perception of an adaptive fin-ray gripper integrating compact deflection sensors," *IEEE Transactions on Robotics*, vol. 39, no. 6, pp. 4482–4499, 2023.

[31] T. Yu, J. Luo, Y. Gong, H. Wang, W. Guo, H. Yu, and G. Chen, "A compact gesture sensing glove for digital twin of hand motion and robot teleoperation," *IEEE Transactions on Industrial Electronics*, 2024.



Tianyi Yan received the B.S. degree in mechanical engineering from Chongqing University, Chongqing, China, in 2022. He is currently working toward the Ph.D degress in mechanical engineering with the Shanghai key Laboratory of Digital Manufacture for Thin-walled Structures, SJTU, Shanghai, China.

His research interests include rigid–flexible coupled mechanism design and motion control.



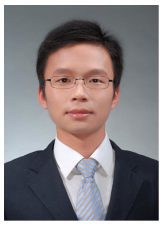
Jianhuan Chen received the B.S. degree in mechanical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2023. He is currently working toward the M.S. degree in mechanical engineering with the Shanghai Key Laboratory of Digital Manufacture for Thin-walled Structures, SJTU.

His research interests include rigid-soft hybrid robot, deflection sensors, and mechanism design.



Siyue Yao received the B.S. degree in mechanical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2022. He is currently working toward to Ph.D. degree in Mechanical Engineering, with both the Shanghai Key Laboratory of Digital Manufacture for Thin-Walled Structures, Shanghai, China and the META Robotics Institute, SJTU.

His research interests include flexible mechanism design, modeling and control, and mechatronics.



Xuyang Duan received the B.S., M.S. and Ph.D. degrees all in mechanical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2010, 2013 and 2024. He is currently a research assistant at the META Robotics Institute, SJTU.

His research interests include the mechanism design and robot control.



Yanjun Wang received his B.S. and M.S. degrees, both in automotive engineering, from Nanhang University, Nanjing, China and Shanghai Jiao Tong University, Shanghai, China, in 2006 and 2009, respectively, and Ph.D. degree in Mechanical Engineering from the University of British Columbia, Vancouver, Canada in 2015. He is currently a research scientist at the Institute of Marine Equipment, SJTU.

His research interests include dynamics modeling of robots, motion planning, force control.



Hao Wang received the B.S., M.S. and Ph.D. degrees all in mechanical engineering from Tianjin University, Tianjing, China, in 1996, 1999 and 2002. He is currently a professor at the Shanghai Key Laboratory of Digital Manufacture for Thin-walled Structures, and the State Key Laboratory of Mechanical System and Vibration, SJTU.

His research interests include the mechanism design, multi-body system dynamics and robotics.



Genliang Chen (Member, IEEE) received B.S. and Ph.D. degrees in mechanical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2006 and 2014, respectively. From 2016 to 2017, he was with the Biomimetic and Dexterous Manipulation Laboratory, Stanford University, as a visiting scholar. He is currently a professor with the META Robotics Institute, and the State Key Laboratory of Mechanical System and Vibration, SJTU.

His research interests include mechanism design, parallel continuum robots, and biomimetic robotics.